

# DSE 2023 Summer School Lausanne

## Lecture 2: Advances in DP Theory

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Loosely based on

- Chapters 8 and 9 of [Dynamic Programming: Foundations](#) by Thomas Sargent and John Stachurski
- [Completely Abstract Dynamic Programming](#) by Thomas Sargent and John Stachurski

Inspired by

- [Abstract Dynamic Programming](#) by Dimitri Bertsekas

# Topics

Handling a large range of dynamic programs

- state-dependent discounting
- recursive preferences
- quantile preferences
- adversarial agents, ambiguity, continuous time, etc., etc.

Generalization  $\implies$  abstraction  $\implies$  clearer proofs

- clarifies optimality conditions
- clarifies relationships between DPs

# Omitted

## Approximation methods

- interpolation
- orthogonal projection
- kernel averages
- neural nets

How do they interact with the algorithms described below?

Some answers in

- Neurodynamic programming (Bertsekas & Tsitsiklis, 1996)
- Subsequent literature
- DP book Vol 2...?

## Prelude: a standard model

Consider a **Markov decision process** (MDP) with

1. a finite set  $X$  called the **state space** and
2. a finite set  $A$  called the **action space**

Actions are restricted by a **feasible correspondence**  $\Gamma$

- from  $X$  to  $A$
- $\Gamma(x) =$  actions available in state  $x$  (nonempty)

- next period state  $x'$  is drawn from  $P(x, a, \cdot)$
- flow **reward** at  $(x, a)$  is  $r(x, a)$
- constant **discount factor**  $\beta \in (0, 1)$

**Lifetime rewards** are

$$\mathbb{E} \sum_{t \geq 0} \beta^t r(X_t, A_t)$$

The **Bellman equation** is

$$v(x) = \max_{a \in \Gamma(x)} \left\{ r(x, a) + \beta \sum_{x'} v(x') P(x, a, x') \right\}$$

A **feasible policy** is a map  $\sigma: X \rightarrow A$  with

$$\sigma(x) \in \Gamma(x) \quad \text{for all } x \in X$$

- $\Sigma :=$  all feasible policies

Feasible policy  $\sigma$  is called ***v-greedy*** if

$$\sigma(x) \in \operatorname{argmax}_{a \in \Gamma(x)} \left\{ r(x, a) + \beta \sum_{x'} v(x') P(x, a, x') \right\} \quad \forall x \in X$$

The **Bellman operator** is

$$(Tv)(x) = \max_{a \in \Gamma(x)} \left\{ r(x, a) + \beta \sum_{x'} v(x') P(x, a, x') \right\}$$

For each  $\sigma \in \Sigma$ , we introduce the **policy operator**

$$(T_\sigma v)(x) = r(x, \sigma(x)) + \beta \sum_{x'} v(x') P(x, \sigma(x), x')$$

Note:

$$\sigma \text{ is } v\text{-greedy} \iff Tv = T_\sigma v$$



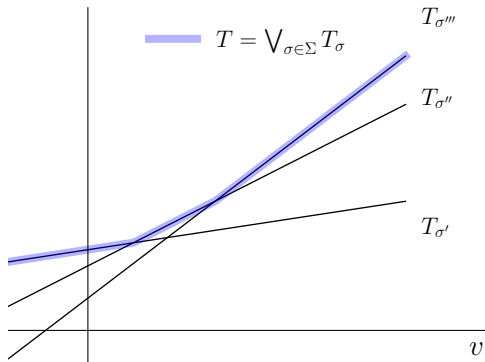


Figure:  $T$  is the pointwise max. of  $\{T_{\sigma}\}_{\sigma \in \Sigma}$

Let

- $r_\sigma(x) := r(x, \sigma(x)) = \text{rewards under } \sigma$
- $P_\sigma(x, x') := P(x, \sigma(x), x') = \text{transitions under } \sigma$

Note that  $P_\sigma$  is Markov dynamics for the state under  $\sigma$

The **lifetime value**  $\sigma$  is

$$v_\sigma(x) := \mathbb{E}_x \sum_{t=0}^{\infty} \beta^t r_\sigma(X_t)$$

where

$(X_t)_{t \geq 0}$  is  $P_\sigma$ -Markov with  $X_0 = x$

Passing the expectation through the sum yields

$$\begin{aligned} v_{\sigma}(x) &= \sum_{t=0}^{\infty} \beta^t \mathbb{E}[r_{\sigma}(X_t) \mid X_0 = x] \\ &= \sum_{t=0}^{\infty} \beta^t \sum_{x'} r_{\sigma}(x') P_{\sigma}^t(x, x') \end{aligned}$$

Using operator / matrix notation, this is

$$\begin{aligned} v_{\sigma} &= \sum_{t \geq 0} (\beta P_{\sigma})^t r_{\sigma} \\ &= (I - \beta P_{\sigma})^{-1} r_{\sigma} \quad (\because \text{Neumann series lemma}) \end{aligned}$$

Recall that the policy operator corresponding to  $\sigma$  is

$$(T_\sigma v)(x) = r(x, \sigma(x)) + \beta \sum_{x'} v(x') P(x, \sigma(x), x')$$

Equivalent:  $T_\sigma v = r_\sigma + \beta P_\sigma v$

Clearly

$$v \in \text{fix}(T_\sigma) \iff v = r_\sigma + \beta P_\sigma v$$

$$\iff (I - \beta P_\sigma)v = r_\sigma$$

$$\iff v = (I - \beta P_\sigma)^{-1} r_\sigma =: v_\sigma$$

**Fact.** :  $T_\sigma^k v \rightarrow v_\sigma$  as  $k \rightarrow \infty$  for all  $v \in \mathbb{R}^X$  ( $\because$  Banach)

# Defining optimality

We define the **value function** via

$$v^*(x) := \max_{\sigma \in \Sigma} v_{\sigma}(x) \quad (x \in \mathbf{X})$$

Equivalently,

$$v^* := \bigvee_{\sigma} v_{\sigma}$$

A policy  $\sigma$  is called **optimal** if  $v_{\sigma} = v^*$

# MDP Optimality

**Theorem.** For an MDP with Bellman operator  $T$  and value function  $v^*$ ,

1.  $v^*$  is the unique fixed point of  $T$  in  $\mathbb{R}^X$
2.  $T$  is a contraction mapping on  $\mathbb{R}^X$
3. A feasible policy is optimal if and only if it is  $v^*$ -greedy
4. At least one optimal policy exists

## Standard algorithms

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**Algorithm 1:** VFI

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input  $v_0 \in \mathbb{R}^X$

input  $\tau$ , a tolerance level for error

$\varepsilon \leftarrow +\infty$

$k \leftarrow 0$

**while**  $\varepsilon > \tau$  **do**

$v_{k+1} \leftarrow Tv_k$

$\varepsilon \leftarrow \|v_k - v_{k+1}\|_\infty$

$k \leftarrow k + 1$

**end**

Compute a  $v_k$ -greedy policy  $\sigma$

**return**  $\sigma$

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**Algorithm 2:** HPI

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input  $\sigma_0 \in \Sigma$ , set  $k \leftarrow 0$  and  $\varepsilon \leftarrow 1$

**while**  $\varepsilon > 0$  **do**

$v_k \leftarrow$  the lifetime value of  $\sigma_k$

$\sigma_{k+1} \leftarrow$  a  $v_k$ -greedy policy

$\varepsilon \leftarrow \mathbb{1}\{\sigma_k \neq \sigma_{k+1}\}$

$k \leftarrow k + 1$

**end**

**return**  $\sigma_k$

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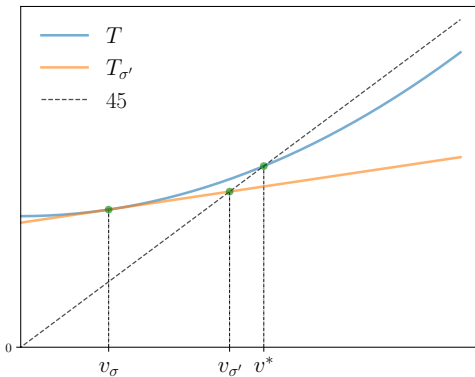


Figure: HPI as a version of Newton's method

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### Algorithm 3: OPI

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input  $v_0$ , an initial guess of  $v^*$

input  $\tau$ , a tolerance level for error

input  $m \in \mathbb{N}$ , a step size

$k \leftarrow 0$

$\varepsilon \leftarrow +\infty$

**while**  $\varepsilon > \tau$  **do**

$\sigma_k \leftarrow$  a  $v_k$ -greedy policy

$v_{k+1} \leftarrow T_{\sigma_k}^m v_k$

$\varepsilon \leftarrow \|v_k - v_{k+1}\|_\infty$

$k \leftarrow k + 1$

**end**

**return**  $\sigma_k$

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**Proposition.** Under the stated condition, VFI, HPI and OPI all converge

Moreover, HPI converges to an exact optimal policy in finitely many steps

For details and proofs see Ch. 5 of <https://dp.quantecon.org/>

# Modifications and extensions

Let's now look at some extensions to the basic model

We can switch to the **expected value function**

$$g(x, a) := \sum_{x'} v(x') P(x, a, x')$$

with “Bellman operator”

$$(Rg)(x, a) = \sum_{x'} \max_{a' \in \Gamma(x')} \{r(x', a') + \beta g(x', a')\} P(x, a, x')$$

- Does  $R$  have the same properties as  $T$ ?
- What are the equivalent algorithms and do they converge?

We can introduce **Epstein–Zin preferences**, as in

$$(Tv)(x) =$$

$$\max_{a \in \Gamma(x)} \left\{ r(x, a)^\alpha + \beta \left( \sum_{x'} v(x')^\gamma P(x, a, x') \right)^{\alpha/\gamma} \right\}^{1/\alpha}$$

- Is  $T$  still a contraction?
- Are the previous optimality results still valid?
- Do VFI, OPI, HPI converge?

We can introduce **risk-sensitive preferences**, as in

$$(Tv)(x) =$$

$$\max_{a \in \Gamma(x)} \left\{ r(x, a) + \beta \frac{1}{\theta} \ln \left( \sum_{x'} \exp(\theta v(x')) P(x, a, x') \right) \right\}$$

- Is  $T$  still a contraction?
- Are the previous optimality results still valid?
- Do VFI, OPI, HPI converge?



We can introduce **risk-sensitive preferences** with **state-dependent discounting**, as in

$$(Tv)(x) = \max_{a \in \Gamma(x)} \left\{ r(x, a) + \beta(x) \frac{1}{\theta} \ln \left( \sum_{x'} \exp(\theta v(x')) P(x, a, x') \right) \right\}$$

- Is  $T$  still a contraction?
- Are the previous optimality results still valid?
- Do VFI, OPI, HPI converge?

Many, many extensions and combinations we can consider

- ambiguity
- expected values in an Epstein–Zin framework
- expected values + ambiguity + state-dependent discounting
- integrated value functions in a risk-sensitive framework in continuous time
- $Q$ -learning, etc., etc.

Is there any unifying theory?

Or are all these problems too diverse?

# Abstraction Level 1: RDPs

1. Construct a DP framework based on an abstraction of the Bellman equation
2. State optimality results in this framework
3. Connect with applications

Builds on work by

- Eric Denardo
- Dimitri Bertsekas
- Takashi Kamihigashi

# Recursive Decision Problems

We begin with a generic version of the Bellman equation:

$$v(x) = \max_{a \in \Gamma(x)} B(x, a, v)$$

- $x \in$  a finite set  $X$  (the state space)
- $a \in$  a finite set  $A$  (the action space)
- $B(x, a, v)$  = total lifetime rewards
  - contingent on current state-action pair  $(x, a)$
  - using  $v$  to evaluate future states

Formally, a **recursive decision process** (RDP) is a triple

$$\mathcal{R} = (\Gamma, V, B), \quad \text{where...}$$

1.  $\Gamma$  is a nonempty correspondence from  $X$  to  $A$

called the **feasible correspondence**

which generates:

- the **feasible state-action pairs**

$$G := \{(x, a) \in X \times A : a \in \Gamma(x)\}$$

- the set of **feasible policies**

$$\Sigma := \{\sigma \in A^X : \sigma(x) \in \Gamma(x) \text{ for all } x \in X\}$$

2.  $V$  is a subset of  $\mathbb{R}^X$  called the **value space**

→ candidates for the value function

3.  $B$  maps  $G \times V$  to  $\mathbb{R}$ , called the **value aggregator**, satisfies

(a) **monotonicity**:

$$v \leq w \implies B(x, a, v) \leq B(x, a, w)$$

(b) **consistency**:

$$x \mapsto B(x, \sigma(x), v) \text{ is in } V \text{ whenever } \sigma \in \Sigma \text{ and } v \in V$$

Example. Every MDP is an RDP

Take  $\Gamma$  as given, set  $V = \mathbb{R}^X$ , and

$$B(x, a, v) = r(x, a) + \beta \sum_{x'} v(x') P(x, a, x')$$

- monotonicity and consistency conditions are trivial to check
- from  $v(x) = \max_{a \in \Gamma(x)} B(x, a, v)$  we recover the MDP Bellman equation

$$v(x) = \max_{a \in \Gamma(x)} \left\{ r(x, a) + \beta \sum_{x'} v(x') P(x, a, x') \right\}$$

**Example.** Consider an **optimal stopping** problem with

$$v(x) = \max \left\{ e(x), c(x) + \beta \sum_{x' \in X} v(x') P(x, x') \right\}$$

Let  $V = \mathbb{R}^X$

If  $\Gamma(x) = \{0, 1\}$  and

$$B(x, a, v) = ae(x) + (1 - a) \left[ c(x) + \beta \sum_{x' \in X} v(x') P(x, x') \right]$$

then  $(\Gamma, V, B)$  is an RDP with the same Bellman equation



**Example.** Consider an MDP with **state-dependent discounting**, so that

$$v(x) = \max_{a \in \Gamma(x)} \left\{ r(x, a) + \sum_{x'} v(x') \beta(x, a, x') P(x, a, x') \right\}$$

Let  $V = \mathbb{R}^X$  and

$$B(x, a, v) = r(x, a) + \sum_{x'} v(x') \beta(x, a, x') P(x, a, x')$$

Now  $(\Gamma, V, B)$  is an RDP with the same Bellman equation

**Example.** Consider a modified MDP with **risk-sensitive preferences**, so that

$$v(x) = \max_{a \in \Gamma(x)} \left\{ r(x, a) + \beta \frac{1}{\theta} \ln \left( \sum_{x'} \exp(\theta v(x')) P(x, a, x') \right) \right\}$$

for nonzero  $\theta$

With  $V = \mathbb{R}^X$  and

$$B(x, a, v) = r(x, a) + \beta \frac{1}{\theta} \ln \left( \sum_{x'} \exp(\theta v(x')) P(x, a, x') \right)$$

we obtain an RDP with the same Bellman equation

**Example.** Consider a modified MDP with **Epstein–Zin preferences**, so that

$$v(x) = \max_{a \in \Gamma(x)} \left\{ r(x, a)^\alpha + \beta \left( \sum_{x'} v(x')^\gamma P(x, a, x') \right)^{\alpha/\gamma} \right\}^{1/\alpha}$$

for nonzero  $\alpha, \gamma$

With  $V =$  the strictly positive functions in  $\mathbb{R}^X$  and

$$B(x, a, v) = \left\{ r(x, a)^\alpha + \beta \left( \sum_{x'} v(x')^\gamma P(x, a, x') \right)^{\alpha/\gamma} \right\}^{1/\alpha}$$

we obtain an RDP with the same Bellman equation

**Example.** Consider a modified MDP with **quantile preferences**, so that

$$v(x) = \max_{a \in \Gamma(x)} \{r(x, a) + \beta(R_\tau^a v)(x)\}$$

where

$$(R_\tau^a v)(x) := \tau\text{-th quantile of } v(X') \text{ when } X' \sim P(x, a, \cdot)$$

With  $V = \mathbb{R}^X$  and

$$B(x, a, v) = r(x, a) + \beta(R_\tau^a v)(x)$$

we obtain an RDP with the same Bellman equation

**Example.** Consider a **shortest path problem** on graph  $\mathcal{G} = (\mathbf{X}, E)$

- $c(x, x') = \text{cost of traversing edge } (x, x') \in E$
- the direct successors of  $x$  denoted by

$$\mathcal{O}(x) := \{x' \in \mathbf{X} : (x, x') \in E\}$$

Aim: find the minimum cost path from  $x$  to a specified vertex  $d$

No discounting (so cannot use MDP theory)

The Bellman equation is

$$v(x) = \min_{x' \in \mathcal{O}(x)} \{c(x, x') + v(x')\}$$

Let  $V = \mathbb{R}^X$

Let  $\Gamma(x) = \mathcal{O}(x)$  and

$$B(x, x', v) = c(x, x') + v(x')$$

This is an RDP with the same Bellman equation

# Policies

Consider an arbitrary RDP  $(\Gamma, V, B)$

A **feasible policy** is a

$\sigma \in A^X$  such that  $\sigma(x) \in \Gamma(x)$  for all  $x \in X$

- respond to state  $X_t$  with action  $\sigma(X_t)$  at **all**  $t \geq 0$
- $\Sigma :=$  the set of all feasible policies

# Policy Operators

Fix  $\sigma \in \Sigma$

The corresponding **policy operator**  $T_\sigma$  is defined at  $v \in V$  by

$$(T_\sigma v)(x) = B(x, \sigma(x), v) \quad (x \in X)$$

**Lemma.**  $T_\sigma$  is an order-preserving self-map on  $V$

Proof: Immediate from monotonicity and consistency



Example. The Epstein–Zin policy operator is

$$(T_{\sigma} v)(x) = \left\{ r(x, \sigma(x))^{\alpha} + \beta \left( \sum_{x'} v(x')^{\gamma} P(x, \sigma(x), x') \right)^{\alpha/\gamma} \right\}^{1/\alpha}$$

# Optimality

To define optimality for RDPs, we use the natural generalizations...

# Lifetime value

Let  $\mathcal{R} := (\Gamma, V, B)$  be an RDP and let  $\sigma$  be any policy

Suppose  $T_\sigma$  has a unique fixed point in  $V$

We denote this function by  $v_\sigma$  and call it the  **$\sigma$ -value function**

We interpret this function as the lifetime value of following  $\sigma$

We call  $\mathcal{R}$  **well-posed** if  $T_\sigma$  has a unique fixed point in  $V$  for all  $\sigma \in \Sigma$

**Example.** Let  $\mathcal{R}$  be the RDP generated by an MDP

Recall that

$$T_{\sigma} v = r_{\sigma} + \beta P_{\sigma} v$$

This operator has the unique fixed point

$$v_{\sigma} = (I - \beta P_{\sigma})^{-1} r_{\sigma}$$

- Hence  $\mathcal{R}$  is well-posed
- $v_{\sigma}(x) = \mathbb{E}_x \sum_{t \geq 0} \beta^t r(X_t, \sigma(X_t)) = \text{lifetime value}$

**Example.** For the Epstein–Zin RDP,

$$(T_{\sigma}v)(x) = \left\{ r(x, \sigma(x))^{\alpha} + \beta \left[ \sum_{x' \in X} v(x')^{\gamma} P(x, \sigma(x), x') \right]^{\alpha/\gamma} \right\}^{1/\alpha}$$

and

$V :=$  the strictly positive functions in  $\mathbb{R}^X$

- Is this RDP well-posed?

# Greedy Policies

Fix  $v \in \mathbb{R}^X$

A policy  $\sigma$  is called  **$v$ -greedy** if

$$\sigma(x) \in \operatorname{argmax}_{a \in \Gamma(x)} B(x, a, v)$$

for all  $x \in X$

Note: at least one  $v$ -greedy policy exists in  $\Sigma$

# The Bellman Operator

The **Bellman operator** is the self-map on  $\mathbb{R}^X$  defined by

$$(Tv)(x) = \max_{a \in \Gamma(x)} B(x, a, v)$$

Key idea:

$$Tv = v \iff v \text{ satisfies the Bellman equation}$$

# Optimality

Let  $\mathcal{R}$  be a well-posed RDP

The **value function** is defined by  $v^* = \bigvee v_\sigma$

More explicitly,

$$v^*(x) := \max_{\sigma \in \Sigma} v_\sigma(x) \quad (x \in X)$$

= max lifetime value from state  $x$

A policy  $\sigma \in \Sigma$  is called **optimal** if

$$v_\sigma = v^*$$



# Howard policy iteration for RDPs

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input  $\sigma_0 \in \Sigma$ , an initial guess of  $\sigma^*$ 
 $k \leftarrow 0$ 
 $\varepsilon \leftarrow 1$ 
while  $\varepsilon > 0$  do
     $v_k \leftarrow$  the unique fixed point of  $T_{\sigma_k}$ 
     $\sigma_{k+1} \leftarrow$  a  $v_k$  greedy policy
     $\varepsilon \leftarrow \mathbb{1}\{\sigma_k \neq \sigma_{k+1}\}$ 
     $k \leftarrow k + 1$ 
end
return  $\sigma_k$ 
```

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Let  $\mathcal{R}$  be an RDP

Key question:

What assumptions to we need for optimality?

Obviously  $\mathcal{R}$  must be well-posed

- each  $T_\sigma$  has a unique fixed point in  $V$

This is the minimum requirement

What else?

# Stability

Let  $\mathcal{R}$  be an RDP

We call  $\mathcal{R}$  **globally stable** if, for all  $\sigma \in \Sigma$ , the operator  $T_\sigma$  is globally stable on  $V$

That is, for all  $\sigma \in \Sigma$ ,

1.  $T_\sigma$  has a unique fixed point  $v_\sigma$  in  $V$  and
2.  $\lim_{k \rightarrow \infty} T_\sigma^k v = v_\sigma$  for all  $v \in V$

Let  $\mathcal{R}$  be a well-posed RDP with value function  $v^*$

**Theorem.** If  $\mathcal{R}$  is globally stable, then

1.  $v^*$  is the unique solution to the Bellman equation in  $\mathbb{R}^X$
2. A feasible policy is optimal if and only if it is  $v^*$ -greedy
3. At least one optimal policy exists
4. HPI returns an optimal policy in finitely many steps
5. VFI and OPI converge

Proof: See Ch 8

# Types of RDPs

The optimality properties require global stability of all  $T_\sigma$

We can check this directly

We can also

1. identify classes of RDPs that are globally stable
2. show that a given application belongs to one of these classes

Let's discuss the classification approach

Below  $\mathcal{R} = (\Gamma, V, B)$  is a fixed RDP

# Contracting RDPs

We call  $\mathcal{R}$  **contracting** if  $\exists \beta < 1$  such that

$$|B(x, a, v) - B(x, a, w)| \leq \beta \|v - w\|_\infty$$

for all  $(x, a) \in G$  and  $v, w \in V$

**Thm.** If  $\mathcal{R}$  is contracting and  $V$  is closed, then  $\mathcal{R}$  is globally stable

Proof: Easy to show that each  $T_\sigma$  is a contraction on  $V$

(Main idea dates back to Denardo 1967)

# Eventually Contracting RDPs

We call  $\mathcal{R}$  **eventually contracting** if there is an  $L \geq 0$  such that  $\rho(L) < 1$  and

$$|B(x, a, v) - B(x, a, w)| \leq \sum_{x'} |v(x') - w(x')| L(x, x')$$

for all  $(x, a) \in G$  and  $v, w \in V$

**Thm.** If  $\mathcal{R}$  is eventually contracting and  $V$  is closed, then  $\mathcal{R}$  is globally stable

Proof: See the book

# Concave RDPs

We call  $\mathcal{R}$  **concave** if

1.  $V = [v_1, v_2]$
2.  $B(x, a, v_1) > v_1(x)$  for all  $(x, a) \in G$  and
3.  $v \mapsto B(x, a, v)$  is concave for all  $(x, a) \in G$

**Thm.** If  $\mathcal{R}$  is concave, then  $\mathcal{R}$  is globally stable

Proof: See the book



# Application: job search with quantile preferences

Set up:

- wage offer process  $(W_t)_{t \geq 0}$  is  $P$ -Markov on finite set  $W$
- discount factor  $\beta \in (0, 1)$

The Bellman equation is

$$v(w) = \max \left\{ \frac{w}{1 - \beta}, c + \beta(R_\tau v)(w) \right\}$$

Here

$$(R_\tau v)(w) := \tau\text{-th quantile of } v(W') \text{ when } W' \sim P(w, \cdot)$$

This problem studied in

- de Castro and Galvao (2019)
- de Castro, Galvao and Nunes (2022)
- de Castro and Galvao (2022)

We embed into the RDP framework by taking

- $\Gamma(w) = \{0, 1\}$
- $V = \mathbb{R}_+^W$
- $B$  given by

$$B(w, a, v) = a \frac{w}{1 - \beta} + (1 - a)[c + \beta(R_\tau v)(w)]$$

Easy to check that  $\mathcal{R} := (\Gamma, V, B)$  is an RDP with Bellman equation

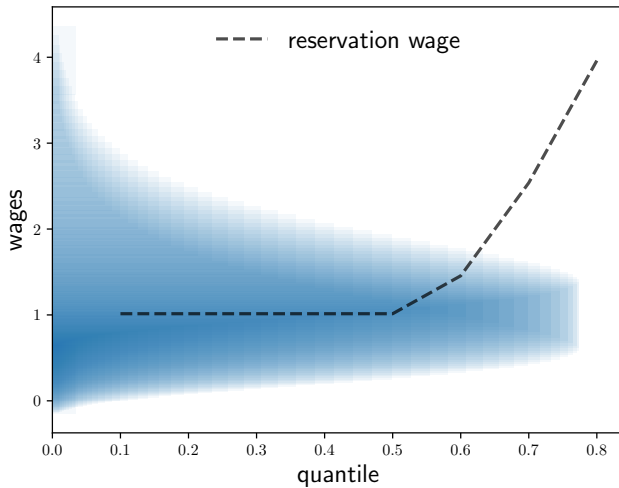
$$v(w) = \max \left\{ \frac{w}{1 - \beta}, c + \beta(R_\tau v)(w) \right\}$$

**Proposition.**  $\mathcal{R}$  is a contracting RDP

- Proof: See DP Ch. 8

Since  $V$  is closed,  $\mathcal{R}$  is globally stable

Hence all optimality properties apply



## Abstraction Level 2: ADPs

We define an **abstract dynamic program** (**ADP**) to be a pair

$$\mathcal{A} = (V, \{T_\sigma\}_{\sigma \in \Sigma}), \quad \text{where}$$

1.  $V = (V, \preceq)$  is a partially ordered set and
2.  $\{T_\sigma\}_{\sigma \in \Sigma}$  is a family of self-maps on  $V$

Below,

- elements of  $\Sigma$  will be referred to as **policies**
- elements of  $\{T_\sigma\}$  are called **policy operators**

If  $T_\sigma$  has a unique fixed point, then we

- denote it  $v_\sigma$
- call it the  **$\sigma$ -value function**

Interpretation:

- $V$  is a set of candidate value functions
- $\Sigma$  is a set of feasible policies
- the lifetime value of  $\sigma \in \Sigma$  is  $v_\sigma$
- we seek a greatest element in  $\{v_\sigma\}_{\sigma \in \Sigma}$

**Example.** Consider an RDP  $(\Gamma, V, B)$

Let  $\Sigma$  be the set of feasible policies

Recall that, for each  $\sigma \in \Sigma$  and  $v \in V$ ,

$$(T_\sigma v)(x) = B(x, \sigma(x), v)$$

The pair  $(V, \{T_\sigma\})$  is an ADP



Recall the expected value Bellman equation

$$g(y, a) =$$

$$\sum_{y'} \int \max_{a' \in \Gamma(y')} \{r(y', \varepsilon', a') + \beta g(y', a')\} P(y, a, y') \varphi(\varepsilon') d\varepsilon'$$

If  $V = \mathbb{R}^G$  and  $R_\sigma$  is defined by

$$(R_\sigma g)(y, a) =$$

$$\sum_{y'} \int \{r(y', \varepsilon', \sigma(y)) + \beta g(y', \sigma(y))\} P(y, a, y') \varphi(\varepsilon') d\varepsilon'$$

then  $(V, \{R_\sigma\})$  is an ADP

## Benefits of ADP theory

- More abstraction means easier proofs
- Removing structure makes it easier to see connections
- Can handle a more diverse range of problems

Given  $v \in V$ , a policy  $\sigma$  in  $\Sigma$  is called  **$v$ -greedy** if

$$T_\sigma v \succeq T_\tau v \quad \text{for all } \tau \in \Sigma \quad (1)$$

**Example.** For an RDP we have

$$(T_\sigma v)(x) = B(x, \sigma(x), v)$$

so (1) holds iff

$$\sigma(x) \in \operatorname{argmax}_{a \in \Gamma(x)} B(x, a, v) \quad \text{for all } x \in X$$

- ADP definitions generalize RDP definitions

# Bellman equation

Fix an ADP  $\mathcal{A} = (V, \{T_\sigma\})$

We define the **Bellman operator** via

$$Tv := \bigvee_{\sigma} T_{\sigma} v$$

(if it exists)

We say that  $v \in V$  satisfies the **Bellman equation** if  $Tv = v$

# Properties

We say that  $\mathcal{A} = (V, \{T_\sigma\})$  is

- **well-posed** if  $T_\sigma$  has one fixed point in  $V$  for each  $\sigma \in \Sigma$
- **order stable** if  $(V, T_\sigma)$  is order stable for each  $\sigma \in \Sigma$
- **max-stable** if  $\mathcal{A}$  is order stable, each  $v \in V$  has at least one greedy policy, and  $T$  has at least one fixed point in  $V$

Note: order stability is a regularity property — see Ch 9

Let  $\mathcal{A}$  be a well-posed ADP

A policy  $\sigma \in \Sigma$  is called **optimal** for  $\mathcal{A}$  if

$$v_\tau \preceq v_\sigma \text{ for all } \tau \in \Sigma$$

We set  $v^* := \bigvee_\sigma v_\sigma$  and call  $v^*$  the **value function**

We define a self-map  $H$  on  $V$  via

$$H v = v_\sigma \quad \text{where } \sigma \text{ is } v\text{-greedy}$$

Iterating with  $H$  is an abstract version of HPI

# Max-Optimality

**Theorem.** If  $\mathcal{A}$  is max-stable, then

1.  $v^*$  exists in  $V$
2.  $v^*$  is the unique solution to the Bellman equation in  $V$
3. a policy is optimal if and only if it is  $v^*$ -greedy
4. at least one optimal policy exists

If, in addition,  $\Sigma$  is finite, then  $\text{HPI} \rightarrow v^*$  in finitely many steps

Proof: See Ch. 9

## Subordinate ADPs

Let  $\mathcal{A} := (V, \{T_\sigma\})$  and  $\hat{\mathcal{A}} := (\hat{V}, \{\hat{T}_\sigma\})$  be ADPs

We say that  $\hat{\mathcal{A}}$  is **subordinate** to  $\mathcal{A}$  if  $\exists$

1. an order-preserving map  $F$  from  $V$  onto  $\hat{V}$  and
2. order-preserving maps  $\{G_\sigma\}_{\sigma \in \Sigma}$  from  $\hat{V}$  to  $V$

such that

$$T_\sigma = G_\sigma \circ F \quad \text{and} \quad \hat{T}_\sigma = F \circ G_\sigma \quad \text{for all } \sigma \in \Sigma$$

Let  $G = \bigvee_\sigma G_\sigma$



**Theorem.** If

1.  $\mathcal{A}$  is max-stable and
2.  $\hat{\mathcal{A}}$  is subordinate to  $\mathcal{A}$ ,

then  $\hat{\mathcal{A}}$  is also max-stable and the Bellman operators are related by

$$T = G \circ F \quad \text{and} \quad \hat{T} = F \circ G$$

while the value functions are related by

$$v^* = G \hat{v}^* \quad \text{and} \quad \hat{v}^* = F v^*$$

Moreover,

1. if  $\sigma$  is optimal for  $\mathcal{A}$ , then  $\sigma$  is optimal for  $\hat{\mathcal{A}}$ , and
2. if  $G_\sigma \hat{v}^* = G \hat{v}^*$ , then  $\sigma$  is optimal for  $\mathcal{A}$

# Application

Consider an Epstein–Zin dynamic program with Bellman equation

$$v(w, e) = \max_{0 \leq s \leq w} \left\{ r(w, s, e)^\alpha + \beta \left( \sum_{e'} v(s, e')^\gamma \varphi(e') \right)^{\alpha/\gamma} \right\}^{1/\alpha}$$

Here

- $w$  is current wealth (discretized)
- $s$  is savings (discretized)
- $e$  is an IID endowment shock with range  $E$
- $\beta$  is a constant in  $(0, 1)$  and  $r$  is a reward function

The policy operator corresponding to  $\sigma \in \Sigma$  is

$$(T_\sigma v)(w, e) = \left\{ r(w, \sigma(w), e)^\alpha + \beta \left( \sum_{e'} v(\sigma(w), e')^\gamma \varphi(e') \right)^{\alpha/\gamma} \right\}^{1/\alpha}$$

**Proposition.** If

- $X := W \times E$  and
- $V := (0, \infty)^X$ ,

then  $\mathcal{A} = (V, \{T_\sigma\})$  is a max-stable ADP

(Details in Ch 9)

Next consider the operator

$$(B_\sigma h)(w) = \left\{ \sum_e \{r(w, \sigma(w), e)^\alpha + \beta h(\sigma(w))^\alpha\}^{\gamma/\alpha} \varphi(e) \right\}^{1/\gamma},$$

where  $h$  is an element of  $(0, \infty)^W$

Define  $F$  at  $v \in V$  by

$$(Fv)(w) = \left\{ \sum_e v(w, e)^\gamma \varphi(e) \right\}^{1/\gamma} \quad (w \in W)$$

Then  $\mathcal{B} = (F(V), \{B_\sigma\})$  is also an ADP

Moreover,  $\mathcal{B}$  is subordinate to  $\mathcal{A}$

To see, this, define  $G_\sigma$  by

$$(G_\sigma h)(w, e) = \{r(w, \sigma(w), e)^\alpha + \beta h(\sigma(w))^\alpha\}^{1/\alpha}$$

Then

- $F$  and  $G_\sigma$  are order-preserving
- $T_\sigma$  is equal to  $G_\sigma \circ F$  and
- $B_\sigma$  is equal to  $F \circ G_\sigma$

---

**Algorithm 4:** Solving  $\mathcal{A}$  via  $\mathcal{B}$ 

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input  $\sigma_0 \in \Sigma$ , set  $k \leftarrow 0$  and  $\varepsilon \leftarrow 1$

**while**  $\varepsilon > 0$  **do**

$h_k \leftarrow$  the fixed point of  $B_{\sigma_k}$

$\sigma_{k+1} \leftarrow$  an  $h_k$ -greedy policy, satisfying

$$\sigma_{k+1}(w) \in \operatorname{argmax}_{0 \leq s \leq w} \left\{ \sum_e \{r(w, s, e)^\alpha + \beta h(s)^\alpha\}^{\gamma/\alpha} \varphi(e) \right\}^{1/\gamma}$$

$\varepsilon \leftarrow \mathbb{1}\{\sigma_k \neq \sigma_{k+1}\}$  and  $k \leftarrow k + 1$

**end**

Compute  $\sigma$  to satisfy

$$\sigma(w, e) \in \operatorname{argmax}_{0 \leq s \leq w} \{r(w, s, e)^\alpha + \beta h_k(s)^\alpha\}^{1/\alpha}$$

**return**  $\sigma$

---

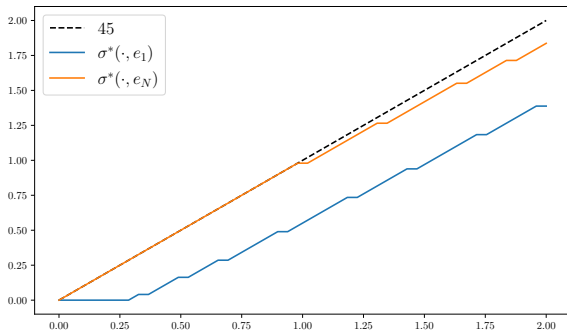


Figure: Optimal savings policy with Epstein–Zin preference

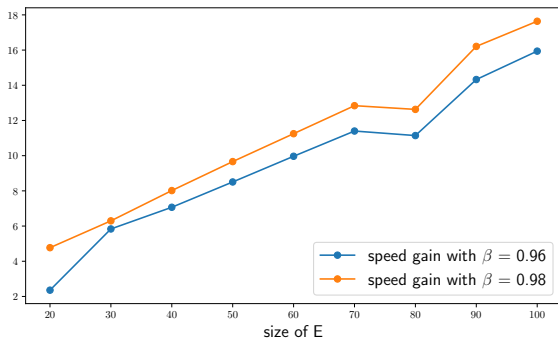


Figure: Speed gain from replacing  $\mathcal{A}$  with subordinate model  $\mathcal{B}$



For details of computations see

[https://github.com/jstac/adps\\_public](https://github.com/jstac/adps_public)